

🎓 EDUCATION

Institute of Atmospheric Physics, Chinese Academy of Sciences 2021 – 2026 (Expected)

Ph.D. in Meteorology Supervisor: Prof. Wansuo Duan

Research Focus: Martian Atmospheric Predictability, Numerical Modeling, Nonlinear Optimization.

UC Berkeley 2020 Spring

Exchange student in Mechanical Engineering

Selected Coursework: Convective Transport and Computational Methods, Nonlinear Dynamics and Chaos.

University of Chinese Academy of Sciences 2017 – 2021

B.Eng. in Theoretical and Applied Mechanics (GPA 3.8/4.0)

Selected Coursework: Thermodynamics and Statistical Physics, Continuum Mechanics, Fluid Mechanics, Introduction to Machine Learning, Introduction to Computational Mechanics, Microgravity Fluid Physics

👤 PROFESSIONAL SUMMARY

Meteorology Ph.D. candidate with a robust background in Applied Mathematics and Planetary Fluid Dynamics. Expert in Martian climate modeling (LMD Mars PCM) and atmospheric predictability, with a focus on the dynamical mechanisms of error growth. Proficient in developing nonlinear optimization algorithms (CNOP) to diagnose atmospheric predictability limits and eager to transition these methodologies to MarsWRF and other interactive models. Seeking to contribute expertise in numerical analysis to Martian weather forecast studies.

📖 PUBLICATIONS & PRESENTATIONS

Peer-Reviewed Publications

1. Yi Zhuang, Wansuo Duan, Li Dong. **A Season-Dependent Weather Predictability Barrier Phenomenon in the Martian Atmosphere.** *Journal of Geophysical Research: Planets*, 131(7), e2025JE009522, 2026. doi:10.1029/2025JE009522.

Selected Conference Presentations

1. **Mars Through Time.** 2025/10. Paris. France.
Poster: Season-dependent Weather Predictability Barrier phenomenon in the Martian atmosphere.
2. **8th Symposium on Nonlinear Atmospheric and Oceanic Science.** 2025/08. Inner Mongolia, China.
Outstanding Poster (Top 6/32): Season-dependent Weather Predictability Barrier phenomenon in the Martian atmosphere.
3. **AOGS-2025.** 2025/07. Singapore.
Oral: Season-dependent Weather Predictability Barrier phenomenon in the Martian atmosphere.

🔍 RESEARCH EXPERIENCES

The Predictability Barrier of Martian Atmosphere and its Mechanism Since 2022/09

Ph.D. Thesis Supervisor: Prof. Wansuo Duan

- By applying Bred Vector (BV) method in the LMD Mars PCM model, we reveal a **season-dependent weather predictability barrier** phenomenon in the Martian atmosphere. We identify a water ice-radiation positive feedback and propose a novel condensation potential index peculiar to Martian atmosphere to uncover its underlying mechanism and origin of its spatiotemporal dependence.
- We build numerical optimization program adapting the Spectral Projection Gradient method to **solve the Conditional Nonlinear Optimal Perturbation (CNOP)** in the LMD Mars model. The CNOP represents initial errors which induce largest forecast error, and indicates predictability limit for Martian atmosphere.

- The CNOP method proves superior in capturing initial atmospheric instabilities, revealing a **novel mechanism involving water ice, dust, and shortwave radiation**. Its further application demonstrates the characteristics of fast-growing initial errors across different sizes and forecast lengths.

Effects of Topography on Nonlinear Internal Wave in Three-layer Flows

2020/09 – 2021/06

Bachelor's Thesis Supervisor: Prof. Zhan Wang

- Through **asymptotic analysis**, we derive governing equations for weakly nonlinear long wave (KdV and KP equation up to 5th order) in a **three-layer shallow water model with topography**. Then, we illustrate numerical results with selected representative topography settings.

Material Transport of Flows induced by Nonlinear Internal Wave

2019/07 – 2020/12

Undergraduate Research Practice Supervisor: Prof. Zhan Wang

- We derived the mathematical expression for **material transport by wave-induced flows**. Then, we explore the characteristics of flows induced by KdV, MCC and DJL type internal waves, and discuss their ability in material transport and contributing factors respectively.

Teaching Assistant, Introduction to Applied Mathematics in Engineering Science

2020 Fall

Organized and refined lecture notes

PROFESSIONAL SKILLS

- **Conduct Experiments with LMD Mars Planetary Climate Model**
 - Develop automation tools using Python and Shell scripting to run massive simulations on Linux HPC clusters, simplifying the workflow from job submission to post-processing.
 - Implement the calculation of Bred Vectors and develop simple assimilation scheme.
 - Develop a CNOP database and Experience-Guided optimization framework for CNOP calculation.
- **Process data and Conduct Numerical Experiments with Python and MATLAB**
 - Obtain and utilize Martian atmospheric data from MCD and EMM SDC.
 - Post-process and visualization, e.g. Longwave Exchange Matrix, Lorenz energy cycle.
 - Explore physical topics numerically, e.g. ENSO recharge oscillator, DJL internal wave simulation.
- **Profound Understanding of Research Fundamentals**
 - Predictability study, origin of forecast errors, concept and application of CNOP.
 - Characteristics and key processes of Martian atmosphere.
 - Normal and stochastic dynamical system, PDF evolution governed by Liouville equation.
 - Mathematical skills for solving ODE/PDE and deriving atmospheric equations.
- Language: English (TOEFL 100 in 2020), Novice French and Japanese.
- Other practiced technical skills: Git, $\LaTeX$, Mathematica (Certified), etc.

TRANSFERRABLE SKILLS

- Managed research group's **multimedia equipment** to ensure seamless hybrid (online/offline) meetings.
- Assisted in **hosting visiting international scholars** 3 times, coordinating their tours and gift exchanges.
- As a member of the **LASG Student Union Presidium**, lead information collection and equipment management for commencement ceremonies and graduate academic forums.
- Participate in **conference logistics** including registration, materials distribution, and poster mounting.

AWARDS AND HONORS

Merit Student of University of Chinese Academy of Sciences (UCAS) 2024/2019	2025/05, 2019/12
Outstanding President of the LASG Student Union Presidium	2024/03
Freshman Scholarship of IAP-CAS	2020/07
Guo Yonghuai Mechanics Honors Scholarship and Competition-Based Scholarship of UCAS	2019/09